Name and surname:______________________________**Student number:** ______________________

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF STATISTICAL SCIENCES - STA2020F: APPLIED STATISTICS

Class Test 4 – 18 May 2026

Time: 100 MINUTES

Max Marks: 40

Internal examiners: Mrs. D. van Wyk-de Ridder, Ms. A Cloete, Ms. G. Carmichael

Instructions

Answer ALL questions in the answer blocks provided. You may **CIRCLE** chosen letter options for multiple-choice and multiple-select type questions.

Please note that you will need access to **RStudio** to complete this test. You need to be prepared to import a dataset in .*csv* format into R and to use the basic commands introduced during your lectures and the RStudio lab sessions/pictorials. You will be required to answer questions about the content covered over the course of the semester and perform calculations by hand.

NB: Round all numeric answers to **TWO DECIMAL PLACES, unless instructed otherwise.**

Where necessary, use the ***as.factor()*** function in R to transform variables to factors.

Make sure to use "." as a **decimal point** and not a comma.

If necessary, increase the level of precision of R output by running: ***options(digits=12)*** before you do any other commands.

Follow the section-specific instructions communicated in the different parts of this test.

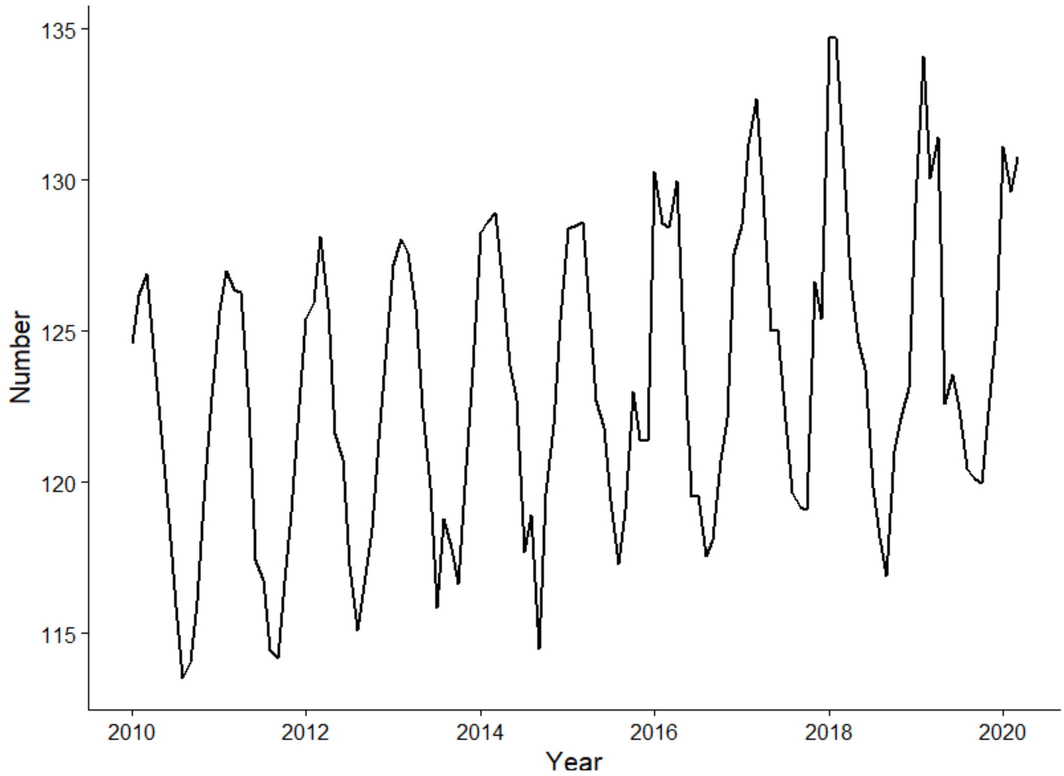
The following files will be downloaded for you by an invigilator. These are the only resources, together with RStudio, that you are allowed to use for this test.

RSTUDIO CHEAT SHEET
FORMULA SHEET
STATISTICAL TABLES
DATASET: bookings.csv
DATASET: student_success.csv
DATASET: metal_data.csv
DATASET: shoe_data.csv

Good luck!

Time Series Section **[15]**

The following time series represents the monthly number of bookings at a coastal guesthouse from January 2010 to March 2020. The time plot of the monthly number of bookings is given below.



Two ARIMA models are fit to the time series of monthly bookings at a coastal guesthouse. The R output is captured below.

ARIMA(1,1,3)

```
Coefficients:
          ar1     ma1     ma2     ma3  intercept
      -0.7884  0.8754  0.0196  0.1442      0e+00
s.e.   0.0777  0.1070  0.1264  0.1170      1e-04

sigma^2 estimated as 3.131e-08:  log likelihood = 958.99,  aic = -1905.99
```

ARIMA(1,1,1)

```
Coefficients:
          ar1      ma1  intercept
      -0.0076  -0.0071      0e+00
s.e.   2.1455   2.1152      1e-04

sigma^2 estimated as 3.443e-08:  log likelihood = 954.05,  aic = -1900.09
```

Instructions:

i) Load the following packages: *tidyverse*, *ggplot2*, *fpp2*, *forecast*, *urca*

ii) Import the file *bookings.csv* into R as an object called *my.data*. Do not open or edit the file in Excel.

iii) Convert the imported values into a time series by using the following command exactly as typed here: ***my.ts <- ts(my.data$value, start = c(2010, 1), frequency = 12).*** This time series will henceforth be referred to as the original time series.

iv) Plot the time series.

v) Construct and interpret the autocorrelation (ACF) and partial autocorrelation (PACF) plots for the original time series.

vi) Perform a classical decomposition on the original time series.

vii) Split the original time series into a training set from the first observation in 2010 until the last observation of 2018, and a test set containing all remaining observations.
Fit
a) the most appropriate simple forecasting method for short-term forecasting, as well as
b) the Holt-Winters method, to the training data,
and compare their forecast accuracy on the test data. Make sure to set your forecasting horizon to the total number of observations in your test dataset.

viii) Use a Box-Cox transformation **and** differencing to remove trend and/or seasonality from the original time series and examine the residuals of your transformed series.

1. Based on the time plot constructed in step (iv) of the monthly number of bookings at a coastal guesthouse, which of the following statements is correct? Circle all the applicable letters. [2]

    A. **The series exhibits an increasing trend together with an annual seasonal pattern.**
    B. The series is stationary because the seasonal pattern repeats regularly over time.
    C. **The time series contains random variation, as evidenced by irregular fluctuations around the trend and seasonal pattern.**
    D. The series exhibits an upward linear trend together with a monthly seasonal pattern.
    E. The variance of the series appears to be converging over time.

2. Carry out step (v). Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F': *The ACF of the original time series shows significant autocorrelation at multiple lags and an oscillating pattern. The PACF, on the other hand, shows significant autocorrelation only at lag 1.* [1]

   FALSE – the statement on the ACF is true and can be observed from the plot, but the PACF also shows significant partial autocorrelations at multiple lags.

3. Why would we be interested in applying moving average smoothing to the monthly number of bookings at the coastal guesthouse? Limit your answer to one sentence. [1]

   Moving-average smoothing removes some of the <u>random variation</u> (1/2) OR <u>seasonality</u> (1/2) to get a clearer understanding of the effects <u>of the components present in the time series</u> (1/2). Also accept <u>"smoothes" the data</u> to get a better idea of the trend (1/2).

4. Carry out step (vi). What is the centred moving average smoothing value (as an estimate for the trend component) for March 2013? Only your final answer will be marked. [1]

   122.26

5. Carry out step (vi). Interpret the adjusted seasonal index for July, which was found to be 0.9643. Limit your interpretation to one sentence. [2]

   An adjusted seasonal index of 0.9643 implies that, <u>on average</u> (1/2), bookings in July are 100*|1-0.9643|=<u>3.57 %</u> (1/2) <u>below</u> (1/2) the <u>annual average</u> bookings (1/2).

6. Investigate the time plot of the monthly number of bookings at a coastal guesthouse from step (iv). Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F': *The most appropriate simple forecasting method for short-term forecasting of this time series is the drift method, as the time series has a trend.* [1]

   FALSE – the time series has strong seasonality, making seasonal naïve more appropriate for short-term forecasting.

7. What is the mean absolute error of the <u>least</u> accurate forecasting method from the methods described in step (vii), based on your R output? Only your final answer will be marked. [2]

   2.25

8. Carry out step (vii). What is the value of α , the smoothing parameter of Holt-Winters' smoothing method, as fit to the training data set? Capture your answer to <u>THREE</u> decimal places. Only your final answer will be marked. [1]

   0.013

9. Based on the R output that you generated in step (viii), which of the following statements are **incorrect**? Circle all the applicable letters. [2]
   A. The p-value for the Ljung-Box test is 0.0003 (rounded to the fourth decimal), indicating that there is autocorrelation present in at least one of the considered lags.
   B. **The plot of the residuals resembles a white noise process.**
   C. **The Box-Cox transformation parameter, λ, is approximately -1, indicating that no variance stabilising transformation is required.**
   D. The residuals are normally distributed, being roughly symmetric and bell-shaped, centred around zero, and not exhibiting skewness or outliers.
   E. **First-order differencing at lag 1 removes seasonality.**

10. Your friend makes the following statement: "*The best fit between the two ARIMA models is the ARIMA(1,1,1) model, as it has the lowest log-likelihood and fewest parameters.*" Do you agree with this statement? Explain your reasoning in a single sentence. [2]

    No (1). The best fit model is the ARIMA(1,1,3) as it has the smallest AIC value (1).

ANOVA Section [10 marks]

A factory produces metal brackets and wants to understand how two process settings affect product strength. Engineers vary Machine Speed (Low, High) and Cooling Method (Air, Water). Thirty-two brackets are manufactured one at a time, with each bracket randomly assigned to one of the four setting combinations (eight brackets per combination). The response is tensile strength (MPa) measured in a standard pull test. This is a 2x2 factorial experiment. The following (incomplete) output in R was obtained.

```
> summary(metal_data)
 MachineSpeed CoolingMethod   Strength
 High:16            Air  :16   Min.   :32.60
 Low :16            Water:16   1st Qu.:37.02
                               Median :42.95
                               Mean   :43.49
                               3rd Qu.:48.92
                               Max.   :61.30
```

```
> model.tables(mod, type = "means")

 MachineSpeed:CoolingMethod
             CoolingMethod
MachineSpeed Air   Water
      High   42.69 42.56
      Low    53.86 34.86
```

```
> model.tables(mod, type = "effects")
Tables of effects

MachineSpeed
  High    Low
-0.8687  0.8687

CoolingMethod
  Air  Water
 4.781 -4.781
```

> summary(mod)

Source	Df	Sum Sq	Mean Sq	F value
Machine Speed	1	24.2	24.2	1.439
Cooling Method	1	731.5	731.5	43.573
Machine Speed × Cooling Method	1	712.5	712.5	42.441
Residuals	28	470.1	16.8	
Total	31	1938.3		

11. What is the interaction effect for the treatment consisting of high speed and air cooling? Only your final answer will be marked. [1]

42.69- (43.49 -0.8687 + 4.781) = -4.7123 = -4.71

12. The Mean Square Residual (MSE) is 16.8. Which of the following statements about this value is correct? Circle the correct letter. [2]

   A. It is the average squared deviation of each observation from its treatment mean and represents the estimated variance of the random error.

   B. It is the variance of the response variable across all 32 brackets.

   C. It is the standard deviation of the residuals, expressed in MPa.

   D. It measures the average difference between the predicted and observed strength values.

   E. It is the sum of squared residuals divided by the total number of observations.

13. Identify what type of an effect −0.8687 is from the output above, and interpret its value in the context of the experiment by referring to the grand mean and including units. [2]

Main effect [1]

The mean tensile strength of brackets produced at High machine speed is 0.8687 MPa below the grand mean [0.5] , averaged across both cooling methods [0.5]. [1]

Or similar to:

On average, the mean tensile strength of brackets produced at high machine speed is 0.87 MPa below the overall mean. [1]

Equivalent phrases for first component:

"The marginal mean at high speed is 0.87 MPa lower than the grand mean"

"Moving from the grand mean to high speed decreases mean tensile strength by 0.87 MPa"

"High speed reduces mean tensile strength by 0.87 MPa relative to the overall average"

"The average tensile strength at high speed is 0.87 MPa less than the grand mean"

Equivalent phrasings for the *averaging* component:

"...averaged over (the levels of) cooling method"

"...marginalising over cooling method"

"...regardless of cooling method"

"...ignoring cooling method"

"...pooled across cooling methods"

14. What is the standard error for interaction effects? Only your final answer will be marked. [1]

sqrt(16.8/8) = 1.45

15. Calculate the p-value associated with the Machine Speed variable using R. Only your final answer will be marked. [1]

0.24

16. The experiment was repeated at a different factory using the same design. Import the *metal_data.csv* dataset into R, generate an interaction plot of Machine Speed and Cooling Method on tensile strength, and circle the letter pertaining to the **most correct** statement based on the plot. [2]

A. Air cooling produces stronger brackets than water cooling, so the factory should switch all production to air cooling.

B. Because the lines cross, there is a statistically significant interaction between Machine Speed and Cooling Method.

C. The plot suggests the effect of Cooling Method depends on Machine Speed; the best setting cannot be chosen without considering both factors together.

D. Since the two cooling methods give roughly the same average strength overall, Cooling Method has no effect on tensile strength.

E. The plot shows that air cooling is better at low speed and water cooling is better at high speed, so the two factors interact.

Non-Parametric Section [6 marks]

A footwear company compares four prototypes of a running shoe: Prototype A, B, C, and D for perceived comfort. A panel of ten trained testers each runs in all four prototypes, with the order of testing randomised for each tester and the testers blinded to the prototype identity. After each run, the tester rates the shoe on a comfort scale from 0 to 100. This gives one score per tester–prototype combination (40 observations in total). The data are stored in *shoe_data.csv*.

17. Which non-parametric test is most appropriate for comparing the four prototypes, and why? Circle the correct letter. [2]

    A. Kruskal-Wallis test, because there are more than two groups to compare.

    B. Friedman test, because each tester rates all four prototypes, so the observations are not independent.

    C. Friedman test, because the comfort scores are not normally distributed.

    D. Kruskal-Wallis test, because the comfort scores are measured on an ordinal scale.

    E. Wilcoxon Signed-Rank test, because the testers are paired.

18. Conduct the appropriate test in R using the built-in function and report the test statistic. Only your final answer will be marked. [1]

    15.60

A medical research team tests whether a new physiotherapy programme reduces lower-back pain. Fifteen patients with chronic lower-back pain each rate their pain on a 0–10 visual analogue scale (VAS) at two time points: once before starting the six-week programme, and once after completing it. Pain scores are bounded and known to be skewed in clinical populations, so the paired Wilcoxon Signed-Rank test is used instead of a paired t-test. The following R output was obtained.

```
> wilcox.test(pain_after, pain_before,
+         paired = TRUE,
+         alternative = "less")

          Wilcoxon signed rank test with continuity correction

data:  pain_after and pain_before
V = 120, p-value = 0.000648
alternative hypothesis: true location shift is less than 0
```

19. For the following question use the R output from *wilcox.test()* to conduct the test and make sure to answer each of the three parts: [3]

- Based on the output, state the null and alternative hypotheses (in words, and without referring to ANY symbols, i.e., not typing 'mu' to represent a mean, etc.).
- Using a 5% significance level, decide whether to reject or not reject the null hypothesis by supplying evidence for your decision, and
- Interpret the decision in the context of the experiment.

Hypotheses:

H0: There is no difference in pain scores before and after the programme. / Pain scores are not lower after the programme. [0.5]

H1: Pain scores are lower after the programme. / The programme reduces pain scores [0.5]

Decision:

Since p = 0.00064 < 0.05 [0.5], we reject the null hypothesis [0.5].

Interpretation:

There is significant evidence that pain scores after the programme are lower than before. [1]

Regression section **[10]**

20. Why do we want to limit the number of independent variables we include in multiple linear regression models? Keep your answer clear and concise. [1]

We lose degrees of freedom on hypothesis tests for each additional variable we include. (1)

OR

To avoid overfitting in the model. (1)

21. If the correlation between *x* and *y* is found to be 0.6, what proportion of the variation in *y* is explained by *x*? Only your final answer will be marked. [1]

0.36

A group of lecturers at the University of Cape Town are conducting a study to understand which factors affect student academic success in the STA2020F course. They record the following information for 30 students who have completed the course and captured the data in the file *student_success.csv*:

- *Mark* – final course mark out of 100
- *Passed* – a binary variable indicating whether a student passed or failed the course (fail=0, pass=1).
- *Sleep* – the number of hours that a student sleeps on average each night.
- *Job* – a binary variable indicating whether a student has a part-time job (no=0, yes=1).
- *Attendance* – the percentage of lectures attended by the student.

Fit a model to determine whether sleep, a part-time job and lecture attendance affect a student's final course mark. Use the results of this model to answer the following questions:

22. Interpret the *Job* coefficient. Be as detailed as possible, while limiting your answer to two sentences. [2]

    On average [0.5], a student's mark decreases by 2.64 [0.5] points/percent when a student has a part-time job compared to when they do not [0.5], holding all else constant [0.5].

23. If a student's attendance percentage increased by 5%, what would be the expected change in their mark? Only your final answer will be marked. [1]
    2.58

24. Report the coefficient of determination for this model (adjusted for the number of independent variables included). Only your final answer will be marked. [1]
    0.86

25. Which of the following statements about the model results are true? Circle all the letters that apply. [2]

    A. The overall model is statistically significant.

    B. The model explains 62.05% of the variability in final course marks.

    C. The model predictions are correct 88% of the time.

    D. On average, marks increase by 3.24 for every additional hour of sleep that a student gets, all else remaining the same.

    E. The degrees of freedom for the tests of significance of the coefficients are 25.

The lecturers realise that they are not actually interested in the exact student course marks. They would rather know whether the independent variables from the analysis above are associated with students passing the course or not. Fit the appropriate model to assist the lecturers in answering this question. Use the results of this model to answer the following questions:

26. Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F': *At the 1% significance level, one would conclude that there is a relationship between sleep and whether a student passes or fails the course*. [1]

    FALSE

27. What is the probability of a student passing if they do not have a part-time job, get 6 hours of sleep each night and attend 50% of lectures? Only your final answer will be marked. [1]

    0.45